Metric Entropy and Observational Entropy Defect

Matched resolutions, hidden fibers, curvature, and heat-trace complexity

Stassis Research Program

Strict GO Core reference revision v0.2 — 2026

Status. This revision is the metric-scale P1 reference module for GO Core v0.2. It replaces dimensional logarithms by reference-scale ratios, uses separate input and output resolutions, types the Lipschitz constant, assigns the heat parameter geometric dimension L^2 before nondimensionalization, and removes the untyped curvature/gradient/entropy sum of v0.1.

Abstract

Covering numbers quantify the number of metric cells needed to resolve a set at a declared scale. Their logarithms are information-valued because their arguments are counts, but all scale limits must use dimensionless ratios. For a Lipschitz observation $O : (X, d_X) \to (Y, d_Y)$, the physically correct interface uses two resolutions, ε_X and ε_Y, satisfying $L_O\varepsilon_X \le \varepsilon_Y$. The matched-scale defect

$$\mathscr{D}^{\mathrm{met}}_{O,b}(A; \varepsilon_X, \varepsilon_Y) = \log_b N_{\varepsilon_X}(A, d_X) - \log_b N_{\varepsilon_Y}(O(A), d_Y)$$

is then nonnegative. Its normalized version has an explicit zero policy and is invariant under changes of length unit and information unit. Defects telescope exactly under composable matched observations. Product fibers, a hidden circle, finite parity, and chromatic graph lifts are treated with stated metric hypotheses. Curvature enters only through dimensionless small-ball asymptotics. The heat-trace logarithm is typed as a smooth spectral complexity proxy, not as Shannon or thermodynamic entropy. The module is a strict interface built from standard metric, Riemannian, and spectral facts; no new gravity law is claimed.

Contents

1 Scope and terminology

This document uses *metric entropy* in the Kolmogorov–Tikhomirov covering sense

$$\mathscr{H}_{b,\varepsilon}(A,d) = \log_b N_\varepsilon(A,d),$$

not in the Kolmogorov–Sinai dynamical sense. The logarithm base $b > 1$ is always declared. A physical metric can be length-valued; an abstract metric may carry another declared scale type. Every set placed inside a logarithm is assumed nonempty. The empty-set covering number is not assigned an entropy in this module.

Boundary 1.1 (Claim boundary). Covering entropy is neither thermodynamic entropy nor physical energy. Curvature influences local volume and hence covering estimates, but this does not derive Einstein's equations. A hidden metric fiber is a mathematical non-identifiability mechanism, not automatically an extra physical dimension.

2 Covering, packing, and scale passports

Definition 2.1 (Internal covering number)**.** Let (X,d) be a metric space, $A \subseteq X$, and $\varepsilon > 0$ with the same scale type as d. Using closed balls $\overline{B}_d(x,\varepsilon)$, define

$$N_\varepsilon(A,d) = \min\left\{|S| : S \subseteq A,\ A \subseteq \bigcup_{s\in S} \overline{B}_d(s,\varepsilon)\right\},$$

and set $N_\varepsilon(A,d) = +\infty$ when no finite cover exists.

Definition 2.2 (Packing number)**.** For $\varepsilon > 0$, define

$$P_\varepsilon(A,d) = \sup\left\{|S| : S \subseteq A,\ d(x,x') > \varepsilon \text{ for } x \neq x'\right\}.$$

For a totally bounded set this number is finite; for compact A the supremum is attained.

Proposition 2.3 (Covering-packing sandwich)**.** *For every totally bounded A and $\varepsilon > 0$,*

$$P_{2\varepsilon}(A,d) \le N_\varepsilon(A,d) \le P_\varepsilon(A,d).$$

Proof. An ε-ball contains at most one point of a 2ε-separated set, which gives the first inequality. A maximal ε-separated set is an ε-net, giving the second. □

Definition 2.4 (Covering entropy)**.** For $N_\varepsilon(A,d) < \infty$,

$$\mathscr{H}_{b,\varepsilon}(A,d) := \log_b N_\varepsilon(A,d).$$

Its physical dimension is **0**, its semantic type is `information`, and its information unit is fixed by b.

2.1 Coherent changes of length and information unit

Theorem 2.5 (Metric-unit covariance)**.** *For every scale factor $a > 0$,*

$$N_{a\varepsilon}(A, ad) = N_\varepsilon(A, d), \qquad \mathscr{H}_{b,a\varepsilon}(A, ad) = \mathscr{H}_{b,\varepsilon}(A, d).$$

Proof. $\overline{B}_{ad}(x, a\varepsilon) = \overline{B}_d(x, \varepsilon)$ as subsets of X, so the admissible covers are identical. □

Theorem 2.6 (Information-unit covariance)**.** *For $b, c > 1$,*

$$\mathscr{H}_{b,\varepsilon}(A, d) = \log_b(c)\, \mathscr{H}_{c,\varepsilon}(A, d).$$

Therefore ratios of finite positive covering entropies computed at the same base are base independent.

3 Dimension without dimensional logarithms

Choose a positive reference scale ℓ_* with the same physical dimension as d, and write

$$\bar{d} = \frac{d}{\ell_*}, \qquad \rho = \frac{\varepsilon}{\ell_*}.$$

Both are dimensionless. The lower and upper box dimensions are

$$\underline{\dim}_B(A, d) = \liminf_{\varepsilon \downarrow 0} \frac{\mathscr{H}_{b,\varepsilon}(A, d)}{\log_b(\ell_*/\varepsilon)}, \tag{1}$$

$$\overline{\dim}_B(A, d) = \limsup_{\varepsilon \downarrow 0} \frac{\mathscr{H}_{b,\varepsilon}(A, d)}{\log_b(\ell_*/\varepsilon)}. \tag{2}$$

When they agree, their common value is $\dim_B(A, d)$.

Proposition 3.1 (Convention independence)**.** *The lower and upper box dimensions are independent of the logarithm base, the coherent length unit, and the choice of fixed positive reference length ℓ_*.*

Proof. Changing the logarithm base multiplies numerator and denominator by the same positive constant. A coherent unit change leaves the ratios d/ℓ_* and ε/ℓ_* unchanged by theorem 2.5. Replacing ℓ_* by $a\ell_*$ adds the constant $\log_b a$ to the denominator; this does not alter the limiting quotient. □

Remark 3.2 (Prohibited shorthand). The expression $\log(1/\varepsilon)$ is not physically typed when ε is a length. It is permitted only after an explicit nondimensionalization, as $\log(1/\rho)$ or $\log(\ell_*/\varepsilon)$.

4 Lipschitz observations at matched resolutions

Let d_X and d_Y have declared scale types U_X and U_Y. A Lipschitz constant has type U_Y/U_X:

$$d_Y(Ox, Ox') \le L_O\, d_X(x, x').$$

When both metrics are lengths, L_O is physically dimensionless.

Definition 4.1 (Matched scale pair)**.** A pair $(\varepsilon_X, \varepsilon_Y)$ is admissible for O if

$$\varepsilon_X > 0, \qquad \varepsilon_Y > 0, \qquad L_O \varepsilon_X \le \varepsilon_Y.$$

Theorem 4.2 (Matched-scale covering contraction)**.** *If $O : (X, d_X) \to (Y, d_Y)$ is L_O-Lipschitz and $(\varepsilon_X, \varepsilon_Y)$ is matched, then for every totally bounded $A \subseteq X$,*

$$N_{\varepsilon_Y}(O(A), d_Y) \leq N_{\varepsilon_X}(A, d_X),$$

and hence

$$\mathscr{H}_{b,\varepsilon_Y}(O(A), d_Y) \leq \mathscr{H}_{b,\varepsilon_X}(A, d_X).$$

Proof. The image of every d_X-ball of radius ε_X is contained in a d_Y-ball of radius $L_O\varepsilon_X \leq \varepsilon_Y$. Map the centers of an optimal finite input cover through O. □

Definition 4.3 (Matched-scale observational defect)**.** For an admissible scale pair, define

$$\mathscr{D}^{\text{met}}_{O,b}(A; \varepsilon_X, \varepsilon_Y) := \mathscr{H}_{b,\varepsilon_X}(A, d_X) - \mathscr{H}_{b,\varepsilon_Y}(O(A), d_Y) \geq 0.$$

This is an information-valued scale profile, not a physical length or energy.

Definition 4.4 (Normalized metric defect)**.** Set

$$\delta^{\text{met}}_O(A; \varepsilon_X, \varepsilon_Y) = \begin{cases} \dfrac{\mathscr{D}^{\text{met}}_{O,b}(A; \varepsilon_X, \varepsilon_Y)}{\mathscr{H}_{b,\varepsilon_X}(A, d_X)}, & \mathscr{H}_{b,\varepsilon_X}(A, d_X) > 0, \\ 0, & \mathscr{H}_{b,\varepsilon_X}(A, d_X) = 0. \end{cases}$$

The second line is the declared one-cell zero policy.

Proposition 4.5 (Invariances and bounds)**.** $0 \leq \delta^{\text{met}}_O \leq 1$*. It is invariant under information-base changes and under coherent rescaling of each metric together with its own resolution.*

Proof. Nonnegativity follows from theorem 4.2; the upper bound uses nonnegativity of the image entropy. Base factors cancel by theorem 2.6, and coherent metric rescaling leaves both covering numbers unchanged by theorem 2.5. □

Theorem 4.6 (Exact composition law)**.** *Let $O_1 : X \to Y$ and $O_2 : Y \to Z$ be Lipschitz. If*

$$L_1\varepsilon_X \leq \varepsilon_Y, \qquad L_2\varepsilon_Y \leq \varepsilon_Z,$$

then

$$\begin{aligned} \mathscr{D}^{\text{met}}_{O_2\circ O_1,b}(A; \varepsilon_X, \varepsilon_Z) = \mathscr{D}^{\text{met}}_{O_1,b}(A; \varepsilon_X, \varepsilon_Y) \\ + \mathscr{D}^{\text{met}}_{O_2,b}(O_1(A); \varepsilon_Y, \varepsilon_Z). \end{aligned}$$

Proof. Expand the two terms on the right. The intermediate covering entropy $\mathscr{H}_{b,\varepsilon_Y}(O_1(A), d_Y)$ cancels. Each term is nonnegative by theorem 4.2. □

Remark 4.7. The uncorrected same-symbol difference $\mathscr{H}_{b,\varepsilon}(A, d_X) - \mathscr{H}_{b,\varepsilon}(O(A), d_Y)$ has no invariant meaning until the input and output scale types and unit conversion are declared.

5 Hidden product fibers

Proposition 5.1 (Product covering bounds)**.** *Let B, F be nonempty totally bounded metric spaces and equip $B \times F$ with*

$$d_\infty((b, f), (b', f')) = \max\left\{d_B(b, b'), d_F(f, f')\right\}.$$

Then

$$N_\varepsilon(B, d_B) \le N_\varepsilon(B \times F, d_\infty) \le N_\varepsilon(B, d_B) N_\varepsilon(F, d_F).$$

For the projection $\pi : B \times F \to B$*,*

$$0 \le \mathscr{D}^{\mathrm{met}}_{\pi,b}(B \times F; \varepsilon, \varepsilon) \le \mathscr{H}_{b,\varepsilon}(F, d_F).$$

Proof. An ε-cover of the product restricts to an ε-cover of the isometric copy $B \times \{f_0\}$, giving the lower bound. The Cartesian product of input covers gives the upper bound. The entropy inequalities follow by taking logarithms. $\square$

Theorem 5.2 (Compact manifold fiber asymptotics)**.** *Let B and F be compact Riemannian manifolds of dimensions m and n, and let a common reference length $\ell_* > 0$ nondimensionalize their distance metrics. For the max product metric,*

$$\mathscr{D}^{\mathrm{met}}_{\pi,b}(B \times F; \varepsilon, \varepsilon) = n \log_b \frac{\ell_*}{\varepsilon} + O(1) \quad (\varepsilon \downarrow 0).$$

Proof. For every compact smooth k-manifold, local Euclidean comparison and a finite atlas yield constants $0 < c < C < \infty$ such that

$$c \left(\frac{\ell_*}{\varepsilon}\right)^k \le N_\varepsilon(M) \le C \left(\frac{\ell_*}{\varepsilon}\right)^k$$

for sufficiently small ε. Apply this to $B \times F$ and B and subtract the logarithms. $\square$

Corollary 5.3 (Hidden circle)**.** *For the intrinsic geodesic metric on a circle S^1_R of radius R,*

$$N_\varepsilon(S^1_R) = \left\lceil \frac{\pi R}{\varepsilon} \right\rceil, \qquad 0 < \varepsilon < \pi R.$$

Hence a forgotten circle contributes

$$\log_b \frac{R}{\varepsilon} + O(1)$$

to the small-scale projection defect.

Proof. A closed geodesic ball of radius ε covers an arc of length 2ε. The circle has circumference $2\pi R$; equally spaced centers attain the stated ceiling. $\square$

Corollary 5.4 (Finite parity fiber)**.** *Let $F = \mathbb{Z}/2\mathbb{Z}$ with point separation $a > 0$. Then*

$$\mathscr{H}_{b,\varepsilon}(F) = \begin{cases} \log_b 2, & 0 < \varepsilon < a, \\ 0, & \varepsilon \ge a. \end{cases}$$

The finite defect therefore depends explicitly on whether the protocol resolves the two points.

6 Operational bridge to probabilistic information

The packing number is the largest codebook size at a declared minimum separation. A uniform random variable X_S on a finite ε-separated codebook S has

$$H_b(X_S) = \log_b |S| \le \log_b P_\varepsilon(A, d).$$

Conversely, a chosen ε-cover produces a deterministic finite quantizer with at most $N_\varepsilon(A, d)$ labels, so its output entropy is at most $\mathscr{H}_{b,\varepsilon}(A, d)$ for every source law. Together with proposition 2.3, this gives an operational packing/covering bracket on finite-resolution distinguishability.

Boundary 6.1 (No automatic equality). The metric defect depends on worst-case covering capacity, while $H_b(X \mid Y)$ depends on a source law and conditional probabilities. They can agree in calibrated finite symmetric examples, but there is no general identity between them.

7 Chromatic graph lifts

Let (X, d_X) be length-valued and let (C, d_C) be a declared feature space. Choose a scale converter $\lambda > 0$ of type $[d_X]/[d_C]$ and define

$$D_\lambda((x,a),(y,c))^2 = d_X(x,y)^2 + \lambda^2 d_C(a,c)^2.$$

For a field $c : X \to C$, its graph is

$$\Gamma_c = \{(x, c(x)) : x \in X\}.$$

Proposition 7.1 (Typed chromatic entropy sandwich). *If c is L_c-Lipschitz, then λL_c is dimensionless and*

$$\mathscr{H}_{b,\varepsilon}(X, d_X) \le \mathscr{H}_{b,\varepsilon}(\Gamma_c, D_\lambda) \le \mathscr{H}_{b,\varepsilon/\sqrt{1+(\lambda L_c)^2}}(X, d_X).$$

Proof. Projection $\Gamma_c \to X$ is 1-Lipschitz. The graph map $x \mapsto (x, c(x))$ is $\sqrt{1+(\lambda L_c)^2}$-Lipschitz. Apply theorem 4.2 to both maps. $\square$

The chromatic excess

$$\mathscr{L}^{\mathrm{ch}}_{b,\varepsilon}(X, c; \lambda) = \mathscr{H}_{b,\varepsilon}(\Gamma_c, D_\lambda) - \mathscr{H}_{b,\varepsilon}(X, d_X)$$

is therefore nonnegative. It measures graph-metric complexity, not hidden probabilistic label entropy. If labels are random conditional on position, the separate quantity $H_b(C \mid X)$ belongs to the information-theoretic module.

8 Local volume, curvature, and a typed pressure model

Let (M, g) be a closed Riemannian n-manifold. Fix $\ell_* > 0$ and define

$$\bar{d} = \frac{d_g}{\ell_*}, \qquad d\bar{V} = \frac{dV_g}{\ell_*^n}, \qquad \overline{\mathrm{Scal}} = \ell_*^2\, \mathrm{Scal}_g, \qquad \bar{r} = \frac{r}{\ell_*}.$$

Every barred quantity in this display is dimensionless.

Proposition 8.1 (Dimensionless small-ball expansion). *As $\bar{r} \downarrow 0$,*

$$\bar{V}(B_{\bar{d}}(x, \bar{r})) = \omega_n \bar{r}^n \left[1 - \frac{\overline{\mathrm{Scal}}(x)}{6(n+2)} \bar{r}^2 + O(\bar{r}^4)\right].$$

Remark 8.2. The statement is local and asymptotic. The sign of scalar curvature controls the leading relative volume correction; it does not by itself determine a finite-scale covering number or a gravitational field equation.

Choose a positive reference information h_* in base-b units and a positive dimensionless reference volume $\bar{v}_*$. A typed local complexity density is

$$\widehat{\rho}_{b,\varepsilon,r}(x) = \frac{\mathscr{H}_{b,\varepsilon}\big(B_g(x,r),d_g\big)}{h_*} \frac{\bar{v}_*}{\bar{V}(B_{\bar{d}}(x,\bar{r}))}.$$

Both ratios are dimensionless and both denominators have strict positivity policies.

Model 8.3 (Dimensionless entropy-pressure functional)**.** Let $\Phi : [0,\infty) \to [0,\infty)$ be declared and let $d\bar{\mu}_g = d\bar{V}/\bar{V}(M)$. Then

$$\mathcal{P}_{b,\varepsilon,r}[g] = \int_M \Phi(\widehat{\rho}_{b,\varepsilon,r}(x))\, d\bar{\mu}_g(x)$$

is dimensionless. Any combination with curvature, gradient, or physical energy terms must first normalize each component separately:

$$\mathcal{J} = \sum_{k=1}^{N} w_k\, \phi_k(\nu_k), \qquad w_k \geq 0, \quad \sum_k w_k = 1,$$

where every ν_k is a declared nonnegative dimensionless quantity with zero and infinity policies. This is a decision functional, not a universal invariant.

9 Heat trace with a geometric scale

Let $\Delta_g \geq 0$ be the nonnegative Laplace–Beltrami operator on a closed Riemannian manifold. Its eigenvalues have physical dimension L^{-2}. Define

$$\overline{\Delta} = \ell_*^2 \Delta_g, \qquad \bar{\lambda}_j = \ell_*^2 \lambda_j, \qquad \tau = \frac{s}{\ell_*^2} > 0,$$

where the geometric heat parameter s has dimension L^2. Then

$$\tau\bar{\lambda}_j = s\lambda_j$$

is dimensionless.

Definition 9.1 (Heat-trace complexity)**.**

$$Z_g(\tau) = \mathrm{Tr}(e^{-\tau\overline{\Delta}}) = \sum_{j=0}^{\infty} e^{-\tau\bar{\lambda}_j}, \qquad \mathcal{C}^{\mathrm{heat}}_{b,\tau}(M,g) = \log_b Z_g(\tau).$$

Boundary 9.2 (Semantic status). $\log_b Z_g(\tau)$ is a log heat trace, analogous to a log partition function. It is not Shannon entropy, von Neumann entropy, or thermodynamic entropy unless a separate normalized state and physical model are supplied.

Theorem 9.3 (Small-τ heat-trace expansion)**.** *For a closed smooth n-manifold,*

$$Z_g(\tau) \sim (4\pi\tau)^{-n/2}\left[\bar{V}(M) + \frac{\tau}{6}\int_M \overline{\mathrm{Scal}}\, d\bar{V} + O(\tau^2)\right] \quad (\tau \downarrow 0).$$

All terms in the bracket are dimensionless.

If t_{phys} is a diffusion time and κ has dimension L^2T^{-1}, then

$$\tau = \frac{\kappa t_{\mathrm{phys}}}{\ell_*^2}$$

is the required dimensionless parameter. Writing $e^{-t_{\mathrm{phys}}\Delta_g}$ without a diffusivity is dimensionally incorrect when Δ_g has dimension L^{-2}.

10 Shell graphs and a conditional cylindrical example

For a finite configuration $P = \{p_1, \ldots, p_N\}$, define a shell graph using a radius $r > 0$ and tolerance $\sigma \geq 0$ of the same dimension:

$$E_{r,\sigma}(P) = \{\{i, j\} : 1 \leq i < j \leq N,\ |d(p_i, p_j) - r| < \sigma\}\,.$$

Under coherent rescaling $(d, r, \sigma) \mapsto (ad, ar, a\sigma)$, the graph is unchanged. Exact unit-distance incidence corresponds to a declared Euclidean reference length and $\sigma = 0$; it is not inferred from covering numbers alone.

Model 10.1 (Cylindrical parameter fiber)**.** Suppose a nondimensionalized compact parameter space has the product form

$$\mathcal{P} = S^2 \times D \times S^1 \times SO(2),$$

where D is a finite set of q separated diameter states. With a regular product metric,

$$\mathscr{H}_{b,\rho}(\mathcal{P}) = 4 \log_b(1/\rho) + \log_b q + O(1).$$

If a Lipschitz observation has bounded image in a one-dimensional interval, then its image covering entropy is at most $\log_b(1/\rho) + O(1)$ at matched dimensionless scales. Thus the defect is at least

$$3 \log_b(1/\rho) + \log_b q + O(1)$$

under these stated hypotheses. No such rate is asserted for an arbitrary non-Lipschitz readout or an undeclared product metric.

11 GO Core quantity and protocol ledger

Quantity	Physical dimension	Semantic type	Required protocol fields
$d_X, \varepsilon_X, \ell_{*,X}$	U_X	metric scale	metric, ball convention, input resolution, reference scale
$d_Y, \varepsilon_Y, \ell_{*,Y}$	U_Y	metric scale	output metric, output resolution, reference scale
L_O	U_Y/U_X	Lipschitz scale converter	domain, codomain, certified or assumed bound
$N_\varepsilon, P_\varepsilon$	$\mathbf{0}$	count	internal/external centers, closed/open balls, separation convention
$\mathscr{H}_{b,\varepsilon}$	$\mathbf{0}$	information in base b	logarithm base and finite/infinite policy
$\mathscr{D}^{\mathrm{met}}_{O,b}$	$\mathbf{0}$	scale-profile information defect	matched pair $(\varepsilon_X, \varepsilon_Y)$
δ^{met}_O	$\mathbf{0}$	normalized loss	positive entropy denominator or one-cell zero policy
s	L^2	geometric heat parameter	ℓ_* or physical diffusivity map
$\tau = s/\ell_*^2$	$\mathbf{0}$	normalized heat scale	strictly positive ℓ_*
λ	$[d_X]/[d_C]$	feature scale converter	feature metric and chromatic protocol

The protocol ledger declares the hidden metric space, system descriptor, reduction (or identity), deterministic observation, stochastic channel (or noiseless statement), observed metric space, reference frame or `not applicable`, quantizer/partition, both spatial resolutions, temporal resolution, observation horizon, estimator/direct readout, uncertainty/exactness status, SI/neutral unit contexts, and defect normalizations.

12 Claim audit and conclusion

Established under stated hypotheses	Not established by this module
Matched Lipschitz observations contract covering numbers.	A same-numerical-scale entropy difference is invariant across arbitrary metric units.
Product fibers contribute their box dimension to logarithmic defect growth.	Every hidden mathematical fiber is a physical compact dimension.
The normalized defect is invariant under coherent length and information unit changes.	Covering entropy is automatically equal to conditional Shannon entropy.
Scalar curvature enters the leading small-ball volume correction.	An entropy-pressure ansatz derives a gravitational field equation.
The nondimensionalized heat trace has standard spectral asymptotics.	The heat-trace logarithm is thermodynamic entropy without a state model.

The strict observable is the scale profile

$$\boxed{(\varepsilon_X, \varepsilon_Y) \longmapsto \mathscr{D}^{\mathrm{met}}_{O,b}(A; \varepsilon_X, \varepsilon_Y), \qquad L_O \varepsilon_X \le \varepsilon_Y.}$$

It is not collapsed into a single universal defect unless a protocol chooses a scale distribution or a decision functional. The principal addition of v0.2 is type safety and compositional closure; the covering, manifold, and heat-kernel results themselves are standard.

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